

Day 3 Lab: Build Something with the Right Architecture

Applied Deep Learning · YDL 2026

Today, we have architectures for image data, sequential data, and graphs. Let's use one of these architectures to take data we're interested in, and use the vectors we can learn from these networks (or the networks themselves) to either A) answer a research question we're interested in, or B) build another intriguing project with the affordances of these networks' embeddings.

The one-sentence brief

Pick a kind of data, embed it with the matching architecture (a CNN for images, an LSTM for sequences, a graph method for molecules or networks), then choose a path: Path A answers a research question with the evidence the network gives you, and Path B builds a creative project on top of what the network learns.

Choose your path

Pick the one that fits what you want to make. Both use the same toolbox and the same data sources, they just point the work in a different direction.

Path A: Answer a research question	Path B: Build a creative project
Use an architecture to investigate something you are curious about. Form a question, then use the network (or the vectors it learns) to gather evidence toward an answer. For example, make a time-series forecast with an RNN/LSTM-informed process, train or fine-tune a model to classify or identify images for a question you care about, or use a GNN to predict a property (a molecular property, or a node or link property in a network).	Use the learned embeddings, or the networks themselves, to build something intriguing. A semantic search or similarity explorer, a recommender, a clustering or map of a collection, or a small interactive app. Same spirit as an open-ended creative project: point the affordances of these networks at a collection you find fun and make something explorable.

Main tasks

You do not need to turn in or work in the architectures toolbox notebook (*lab_architectures_toolbox.ipynb*), but it has runnable, GPU-ready examples for every tool below, and it is built for Google Colab.

- Use at least one architecture from this week on data you gathered yourself.
- Show it works end to end: turn your data into vectors, then either answer your question (Path A) or build your project (Path B).
- Produce one artifact that tells a story: a labelled plot, an interactive map, a forecast, or a short finding about what you saw.
- Building a small local app from your embeddings is an option, not a requirement.

Your toolbox (in the notebook)

- **Images to vectors:** a pretrained CNN as a feature extractor, plus Grad-CAM to see what it looks at.
- **Transfer learning:** fine-tune a pretrained model on your own labels, for the CNN and for the LSTM.
- **Sequences to vectors:** a character-level LSTM encoder that summarizes a string or a series.
- **Molecules to vectors:** RDKit fingerprints and a quick property predictor.
- **Networks:** NetworkX analysis and node embeddings (node2vec) for social graphs.
- **Explore and export:** nearest neighbors, clustering, a UMAP map, and saving your vectors to use locally.

Everything runs on Colab with a free GPU, and every loop shows a tqdm progress bar so you can see runtimes. Use Claude freely for visualization, unique aesthetics, and the optional app.

Project ideas (mix and match)

Each idea names the tools it combines and spans the three architectures (CNN for images, RNN/LSTM for sequences, GNN or graph methods for molecules and networks). **These are just suggestions, feel free to do your own thing, and own idea, so long as it makes use of one of the architectures from this week.** Treat these as starting points, and bend them toward whatever you find fun.

Path A: answer a research question

Question	What it is	Combines
Does my CNN see what I think	Fine-tune or run a pretrained CNN on your own labelled images, then use Grad-CAM to ask where it looks and collect the cases where it is confidently wrong.	CNN + transfer learning + Grad-CAM
Custom image classifier	Train a model on a small set of your own labelled images (your pets, plants, or satellite tiles) to answer a classification question, and report what it learned.	CNN + transfer learning
Can I forecast this series	Embed sequences of weather, energy, or your own tracked numbers with an LSTM-informed process, then forecast ahead and check how well the pattern holds.	LSTM sequence vectors + forecast
Predict a molecular property	Featurize molecules with RDKit fingerprints (or a GNN) and predict a property such as solubility or toxicity on a small public set, then ask which features drove it.	GNN / graph fingerprints + a model
Predict a node or link property	Take a social or citation graph, embed the nodes, and predict a node label or a missing link, then ask what structure the model relied on.	GNN / node2vec + a predictor

Path B: build a creative project

Idea	What it is	Combines
Image map or search engine	Embed a set of images with a pretrained CNN, then build a 2D map or a similarity search over them. Wrap it in the interactive viz, or the optional local app.	CNN image2vec + UMAP + search
Style generator or text encoder	Train a character LSTM on a corpus you love (lyrics, scripts, an author), then generate text in that style or embed and compare passages.	LSTM + fine-tuning

Idea	What it is	Combines
Sequence pattern map	Embed sequences of weather, energy, or your own tracked numbers with an LSTM, then cluster the patterns into an explorable map.	LSTM sequence vectors + clustering
Social-network explorer	Take a social or citation graph, find communities and influential nodes with NetworkX, and embed the nodes to map who is similar to whom.	NetworkX + node2vec
Embed anything recommender	Pick a domain, embed its items with the matching architecture, and build a nearest-neighbor recommender or an explorable map.	Any architecture + the explore toolkit

Where to find data

A few reliable starting points by data type. Always skim a source's terms before redistributing, and prefer openly licensed data.

Images

Source	What / how to access	Notes
torchvision datasets	One line: CIFAR-10, Fashion-MNIST, Oxford Pets, EuroSAT satellite tiles	Easiest; great for fine-tuning
Your own photos	A folder of images you care about	Most personal projects start here
Kaggle / Hugging Face	Thousands of image sets (kaggle.com/datasets, huggingface.co/datasets)	Filter by topic and license

Sequences and text

Source	What / how to access	Notes
Project Gutenberg	Public-domain books via the Gutendex API (gutendex.com)	Great for a style generator
lyrics.ovh	Free lyrics API, no key (api.lyrics.ovh)	Fun corpora for an LSTM
Time series	Weather, energy, or finance series; UCI and Kaggle have many	Or your own tracked numbers

Molecules and networks

Source	What / how to access	Notes
MoleculeNet	ESOL solubility, Tox21 toxicity, BBBP; CSVs of SMILES and labels	Featurize with RDKit
PubChem / ChEMBL	Huge public chemistry databases	For a bigger molecule project

Source	What / how to access	Notes
NetworkX built-ins	karate_club_graph, Les Miserables, and more	Zero setup for a graph project
SNAP / Cora	Stanford SNAP datasets; the Cora citation graph	Real social and citation networks

Exporting from Colab

Your vectors live on the Colab machine. To build on your own computer, save them and download them. The toolbox shows both routes:

- **Quick download:** from `google.colab import files;`
`files.download('embeddings.npy')` sends the file to your local Downloads folder.
- **Larger files:** mount Google Drive with `from google.colab import drive;`
`drive.mount('/content/drive')`, then copy your files into `/content/drive/MyDrive/`.
- **Optional app:** the toolbox writes a small `app.py`; download it next to your embeddings, then run `streamlit run app.py` locally for a similarity search. Building an app is optional on either path.

Tips

Start small, get something working on a slice of data, then scale up. On Path A, decide up front what evidence would answer your question; on Path B, get one thing explorable early. Lean on the toolbox examples for the implementation, and use Colab's free GPU. For visualization, unique aesthetics, and the optional app, feel free to use Claude.

Toolbox libraries: PyTorch and torchvision, pytorch-grad-cam, RDKit, NetworkX, node2vec, scikit-learn, umap-learn, tqdm, and optionally Streamlit and DeepChem.